

The Roof and the Clock

Rowmotion on the E_7 minuscule poset, its linearity across the minuscule family, and the Weyl group it does not share

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Abstract

Let P be a simply-laced minuscule poset and R rowmotion on its order ideals $J(P)$, acting on the weights $W\lambda$ of the minuscule representation. By a theorem of Rush and Shi, R is conjugate to a Coxeter element; a Coxeter element is an isometry of the weights and R in general is not. We prove that R is itself a Weyl group element exactly when P is a chain, that its half-turn $R^{h/2}$ is a Weyl group element exactly when P is a chain or λ is the vector representation of D_n , and we give in closed form, on two families, the fraction of weight pairs whose inner product R preserves. On the 56 weights of E_7 , a pair keeps its inner product under one step of R if and only if a parity computed from the two toggle sets is even. We then determine what a Weyl group shares with its own translates by R : $W \cap R^k W R^{-k}$ equals the centralizer $C_W(R^k)$ on every board of rank at most 7 on which no power of R is linear, with one exception, the D_5 half-spin (the B_4 spinor board), whose mechanism we prove; on the spinor boards B_4 , B_8 and B_{16} the half-turn agrees with a Weyl element of one fixed shape on the middle-rank orbits, and we prove that this element commutes with the half-turn — the property behind the exception — if and only if $n = 4$. The longest element always reverses R , and three label statistics are homomesic. Every computational statement is reproducible from the listed scripts.

Keywords. Rowmotion; minuscule posets; Coxeter elements; Weyl groups; homomesy; the 56-dimensional representation of E_7 ; bitangents of plane quartics; the Fano plane.

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1. Introduction

1.1 Rowmotion and the Coxeter element

For a finite poset P , *rowmotion* is the bijection of the set $J(P)$ of order ideals

$$R(I) = \text{the order ideal generated by the minimal elements of } P \setminus I.$$

Its study goes back to Brouwer–Schrijver, Fon-der-Flaass and Cameron–Fon-der-Flaass [CF95]; the modern theory is surveyed in [SW12, Hop24]. When $P = P_\lambda$ is the *minuscule poset* of a minuscule weight λ of a simple Lie algebra — the poset of join-irreducibles of the distributive lattice formed by the weights of V_λ under the root order [Pro84, Ste94, Gre13] — Rush and Shi proved that rowmotion is conjugate, under Stembridge’s bijection $J(P_\lambda) \cong W/W_\lambda$ [Ste01], to the action of any Coxeter element c of the Weyl group W [RS13]. In particular R has order the Coxeter number h , its orbit structure is that of c , and $(J(P_\lambda), R)$ exhibits cyclic sieving [RSW04].

A Coxeter element acts on the weights $W\lambda$ as an isometry. Rowmotion, carried to $W\lambda$ by the same bijection, is a permutation of the weights that is conjugate to c in the *symmetric group of the weights*, but it need not be an isometry, and in general it is not even a Weyl group element. The first question of this paper is: **for which minuscule posets is rowmotion, or a power of rowmotion, itself an element of W , and how far from an isometry is it otherwise?**

1.2 Results

Throughout, λ is a simply-laced minuscule weight (ω_k of A_n ; $\omega_1, \omega_{n-1}, \omega_n$ of D_n ; ω_1, ω_6 of E_6 ; ω_7 of E_7), $P = P_\lambda$, h the Coxeter number, W the Weyl group acting on the weights $W\lambda$, and R rowmotion on $W\lambda$. Inner products are normalised by $\langle \alpha, \alpha \rangle = 2$ for roots.

Theorem A (linearity). $R \in W$ if and only if P is a chain, i.e. $\lambda \in \{\omega_1, \omega_n\}$ of A_n . If h is even, $R^{h/2} \in W$ if and only if P is a chain or λ is the vector representation ω_1 of D_n ($n \geq 3$, with $D_3\omega_1 = A_3\omega_2$ and, for D_4 , its triality images); in the vector case R^{n-1} is the signed permutation $e_i \mapsto -e_{n-i}$ ($i < n$), $e_n \mapsto (-1)^{n-1}e_n$. More generally, if $R^j \in W$ then j is a period of the extended rank sequence of P (§3).

Theorem B (the defect). Let $\kappa(R)$ be the fraction of unordered pairs of weights whose inner product R preserves. Then for $n \geq 3$

$$\kappa(D_n \omega_1) = 1 - \frac{2(n-1)}{n(2n-1)}, \quad \kappa(A_n \omega_2) = 1 - \frac{3n^2 - 9n + 4}{\binom{n+1}{2}}.$$

Table 1 gives κ , the random baseline, and the Hamming distance from R to the nearest Weyl element for the 42 simply-laced minuscule cases of rank ≤ 7 .

Theorem C (the parity rule on E_7). Let Λ be the 56 weights of the minuscule representation of E_7 , B the inner product reduced mod 2, and for $u \in \Lambda$ let $T(u)$ be the multiset of colours toggled by R at u . For a non-antipodal pair $\{u, u'\}$ let t_1 be the number of colours in $T(u)$ at which u' has a nonzero label, t_2 the same with the roles exchanged, and t_3 the number of Dynkin-adjacent pairs between $T(u)$ and $T(u')$. Then

$$B(Ru, Ru') = B(u, u') \iff t_1 + t_2 + t_3 \equiv 0 \pmod{2}.$$

The rule holds with the k -fold toggle multisets for every power R^k . On E_7 , $\kappa(R) = 501/770$, and no power of R other than the identity is a Weyl element.

The second question concerns what the symmetry group shares with the clock. Write

$$I_k := W \cap R^k W R^{-k}, \quad C_k := C_W(R^k),$$

so that I_k is the set of Weyl elements that are still Weyl elements after conjugation by R^k (“the grammar two instants share”) and $C_k \subseteq I_k$.

Theorem D (reversal). On every simply-laced minuscule board the longest element w_0 reverses rowmotion: $w_0 R w_0 = R^{-1}$. The Weyl elements reversing R form the coset $w_0 C_W(R)$; on E_7 the reverser is unique and equals -1 . On a self-dual board R^k preserves exactly $\frac{1}{2} \text{fix}(R^{2k})$ antipodal pairs.

Theorem E (the shared grammar). (i) On E_7 , $I_k = \{1\}$ for $k \neq 9$ and $I_9 = \{\pm 1\}$; moreover $\langle W, R \rangle = A_{56}$. (ii) On the vector boards $D_n \omega_1$, $I_k = C_W(R^{2k})$ for every k ; more generally, if an involution ι centralizes W and reverses R , then $I_k \subseteq C_W(R^{2k})$. (iii) On the 23 boards of rank ≤ 7 on which no power of R is a Weyl element, $I_k = C_k$ for every k , with one exception, the two half-spins of D_5 (equivalently the B_4 spinor board, the vertices of the 4-cube), where $|I_4| = 8 > |C_4| = 4$ and $|I_3| = |I_5| = 2 > 1$.

Theorem F (the 4-cube). On the B_4 spinor board R has two free orbits $O_1 \ni \lambda$ and O_2 . The half-turn R^4 agrees with the Weyl element $-\tau$, $\tau = (e_2 e_3)(-e_4)$, on all of O_2 and with no Weyl element on O_1 ; consequently $I_4 = C_W((-\tau)R^4)$, of order 8. The obstruction on O_1 is that O_1 is the full-height orbit: for any Weyl element $w = (\pi, v)$ of the hyperoctahedral group the rank-shift is $\text{rank}(wx) - \text{rank}(x) = \text{wt}(u) - 2|x \cap u|$ with $u = \pi^{-1}v$, and an orbit containing λ and spanning $\mathbb{F}_2^n$ cannot carry a constant shift magnitude ≥ 2 .

The phenomenon does not recur on B_8 , the next board on which the clock acts freely: there the half-turn again agrees with a Weyl element on exactly two orbits, but the correction’s centralizer is $\{\pm 1\}$ and $I_k = C_W(R^k)$ at every lag (Proposition 6.7). The reason is a commutation (Lemma 6.9): the Weyl element nearest to the half-turn

commutes with it on B_4 and on no other enumerable spinor board (Proposition 6.10); on B_{16} , computed at the orbit level, the same element again fails to commute and the correction's centralizer is again $\{\pm 1\}$ (Proposition 6.11). The last of these is a theorem for every even n :

Theorem G (why four). *Let $n \geq 4$ be even and let $w_n = -(e_2 e_3)(e_4 e_5) \cdots (e_{n-2} e_{n-1})$ be the signed permutation negating $e_1, \dots, e_{n-1}$, exchanging e_{2i} and e_{2i+1} , and fixing e_n (the notation is fixed in §6.4); for $n = 4, 8, 16$ this is the element that agrees with the half-turn R^n on the middle-rank orbits of the B_n spinor board (Theorem F, Propositions 6.7 and 6.11). Then w_n commutes with R^n if and only if $n = 4$. Consequently the correction $w_n^{-1} R^n$ is an involution only at $n = 4$, and the mechanism of Theorem F cannot enlarge the shared grammar on the agreement set of w_n for any even $n \geq 6$. The proof (§6.4) rests on a closed form for rowmotion on the shifted staircase in excess coordinates and for the orbit of λ , which also shows that the half-turn shifts every rank on that orbit by exactly $\pm n/2$ for every even n ; the two kinds of orbit on which commutation is forced — the orbit of λ and the orbits carrying the half-turn — exhaust the board exactly at $n = 4$, and for $n \geq 6$ the weight with a single minus sign at position 2 is an explicit witness.*

We also show (§7) that the three ternary label counts on any simply-laced minuscule board are homomesic under R with means $|P|/h$, $|P|/h$, $r - 2|P|/h$ — Rush–Wang's antichain homomesy [RW15] read on the Dynkin labels — and that the toggle size $|I \triangle R(I)|$ is homomesic with mean $2|P|/h$, a corollary of Defant–Hopkins–Poznanović–Propp [DHPP23].

The third part of the paper (§§8–9) is the explicit model in which these objects were first met. Two finite groups built independently — a chain inside $PSL(2, 7)$ and a tower of double covers — $SL(2, 7)$, $2 \cdot I$, $GL(2, 3)$, $2 \cdot W(E_6)$, $2 \times PGL(2, 7)$ — are exhibited as subgroups of the rotation subgroup $W^+(E_8)$ over a single central C_2 ; the quotient $O_8^+(2)$, its triality automorphism as an explicit permutation Φ of order 3 on 360 points, its fixed group $G_2(2)$ as an exact intersection, and the fourteen outer conjugacy classes of $O_8^+(2):3$ are all constructed in one coordinate system and matched class by class to the ATLAS. The 56 weights of E_7 are the vectors u with $B(u, v) = 1$ for a fixed nonsingular v ; -1 is the transvection t_v ; and the involution

$$\text{pr} := \iota \circ w_0(E_6) \circ t,$$

where t swaps the two “poles” of the $\omega_7^\vee$ -grading, is the order isomorphism between the two 27-sheets' ideal lattices induced by the E_6 diagram automorphism. Finally $\langle R, \text{pr} \rangle = S_{56}$. Inside $Sp_6(2) \cong W(E_7)/\{\pm 1\}$ acting on the 28 bitangents of a plane quartic, $W(E_6)$ is the stabilizer of a bitangent, $PSL(2, 7)$ is transitive on all 28, and their intersection is S_3 ; the 36 even theta characteristics fall into $PSL(2, 7)$ -orbits of sizes 1, 7, 7, 21, the two orbits of size 7 being the points and the lines of one Fano plane, each realising the 6-dimensional irreducible χ_6 as its deleted permutation module.

1.3 What is classical, what is new, what is computed

The Coxeter-number periodicity of R , the structure of $O_8^+(2):3$, the fixed group of triality, the action of the E_6 diagram automorphism on minuscule posets, and the bitangent geometry of $Sp_6(2)$ are classical and are cited at first use. Theorems A–G and Theorem C's rule are, to the best of a documented search (Appendix A), not in the literature; we claim them as *not found* rather than as new, and would welcome a reference. Statements marked **(C)** were established by exhaustive computation; the scripts, their check counts and their input dependencies are listed in Appendix B, and each was executed independently by both authors on separate machines with identical tallies. Conjectures that we tested and found false are recorded as Remarks where the failure is informative.

1.4 Organisation

§2 fixes notation. §3 proves Theorems A and B. §4 treats the E_7 board: the parity rule, the shared grammar on the 56, and generation of S_{56} . §5 proves Theorem D and the vector-board part of Theorem E and reports the family computation. §6 proves Theorem F, describes the chain $P_j(k)$ of iterated shared grammars, reports the B_8 and B_{16} computations, and proves Theorem G. §7 is homomesy. §8 describes the computational model of $W^+(E_8)$, $O_8^+(2):3$ and $G_2(2)$ and identifies the 56, the transvection and the mirror. §9 is the bitangent bridge

and the Fano plane. §10 lists open problems. Appendix A is the literature search; Appendix B the verification table.

2. Preliminaries

Weights and posets. For a minuscule weight λ the weights of V_λ form the single orbit $W\lambda$; each weight w has Dynkin labels $\langle w, \alpha_i^\vee \rangle \in \{-1, 0, 1\}$. The root order $w \leq w'$ iff $w' - w \in \mathbb{N}\{\alpha_i\}$ makes $W\lambda$ a distributive lattice with top λ and bottom $w_0\lambda = -\lambda^*$; its join-irreducibles form the minuscule poset $P = P_\lambda$, coloured by simple roots, and the ideals $J(P)$ correspond to the weights, $\lambda \leftrightarrow P$, $w_0\lambda \leftrightarrow \emptyset$, with $\$w(I) - w(I') = \$$ the sum of the colours of $I' \setminus I$ for $I \subseteq I'$ [Pro84, Ste94]. The multiplicity c_i of colour i in P is the coefficient of α_i in $\lambda - w_0\lambda = \lambda + \lambda^*$. P is graded; we write s_j for the number of elements of rank j , $0 \leq j \leq h - 2$.

Rowmotion on weights. Under $J(P) \cong W\lambda$, R sends the weight of I to the weight of $\downarrow \min(P \setminus I)$. The multiset $T(w)$ of colours of the elements of $I \triangle R(I)$ is the *toggle set* at w ; the weight changes by the signed sum of the corresponding simple roots. **Theorem (Rush–Shi [RS13]).** R is conjugate in the toggle group to the product of the toggles in any linear order, and under Stembridge’s bijection that product corresponds to a Coxeter element; hence R has order h and the cycle type of a Coxeter element on $W\lambda$.

Boards. We call the pair $(W \curvearrowright W\lambda, R)$ a *board*; it is *rich* when no power of R is a Weyl element and $C_W(R) = 1$. A board is *self-dual* when -1 permutes $W\lambda$ (i.e. $\lambda^* = \lambda$); then the *antipode* $\iota : w \mapsto -w$ is a permutation of the weights, equal to w_0 when $-1 \in W$ (D_n , n even; E_7) and not a Weyl element otherwise ($A_{2k-1}\omega_k$, $D_n\omega_1$ with n odd).

The hyperoctahedral case. The spinor representation of B_n is minuscule with weights $(\pm \frac{1}{2})^n$; we read a weight as $x \in \mathbb{F}_2^n$ (bit $i = 1$ iff coordinate i is negative), so $\text{rank}(x) = \text{wt}(x)$, $\lambda = 0$, $w_0\lambda = \mathbf{1}$. Its poset is the shifted staircase δ_n and $W(B_n) = \mathbb{F}_2^n \rtimes S_n$ acts affinely, $w(x) = \pi(x) \oplus v$. The D_{n+1} half-spin has the same poset and the same rowmotion, with $W(B_n) < W(D_{n+1})$.

Notation for the shared grammar. For a board and $1 \leq k \leq h - 1$, $I_k = W \cap R^k W R^{-k}$ and $C_k = C_W(R^k)$; the *transport* is the map $g \mapsto R^{-k} g R^k$ on I_k (which by definition lands in W). Conjugating by R^k shows $|I_k| = |I_{-k}|$.

3. Linearity of rowmotion

Let $\sigma = (s_0, s_1, \dots, s_{h-2}, c_k)$ for $\lambda = \omega_k$; we call σ , of length h , the *extended rank sequence*, indices taken mod h .

Lemma 3.1. (a) $R(\lambda) = w_0\lambda$; the bottom weight $w_0\lambda$ has a single nonzero label, -1 at the dual node k^* , hence a single cover $p_0 = w_0\lambda + \alpha_{k^*}$, the unique minimal element of P , and $R(w_0\lambda) = p_0$. Consequently

$$\langle \lambda, w_0\lambda \rangle = \langle \lambda, \lambda \rangle - c_k, \quad \langle w_0\lambda, p_0 \rangle = \langle \lambda, \lambda \rangle - 1.$$

(b) The R -orbit of $w_0\lambda$ is the sequence of rank truncations $P_{<j}$, $j = 0, \dots, h - 1$; writing $w(j)$ for their weights,

$$\langle w(j), w(j+1) \rangle = \langle \lambda, \lambda \rangle - \sigma_j \quad (j \bmod h).$$

Proof. (a) $R(P) = \downarrow \min(\emptyset) = \emptyset$ and $R(\emptyset) = \downarrow \min P = \{p_0\}$. Since $\lambda - w_0\lambda = \sum_i c_i \alpha_i$ and $\langle \omega_k, \alpha_i \rangle = \delta_{ik}$, $\langle \lambda, \lambda - w_0\lambda \rangle = c_k$; and $\langle w_0\lambda, \alpha_{k^*} \rangle = -1$ because that is the label of $w_0\lambda$ at k^* . (b) The minimal elements of $P \setminus P_{<j}$ are the rank- j elements, which generate $P_{<j+1}$; so $w(j+1) - w(j)$ is the sum of the colours of the rank- j elements, each added at a label -1 of $w(j)$, and the inner product drops by s_j ; all weights have the same norm. The wrap-around step $\lambda \rightarrow w_0\lambda$ drops by c_k by (a). $\square$

Theorem 3.2 (= Theorem A, first part). $R \in W \iff P$ is a chain $\iff \lambda \in \{\omega_1, \omega_n\}$ of A_n ; then R is the Coxeter element $s_1 \cdots s_n$, the cyclic shift of $e_1, \dots, e_{n+1}$.

Proof. If $R \in W$ it is an isometry, so applying it to the pair $(\lambda, w_0\lambda) \mapsto (w_0\lambda, p_0)$ of Lemma 3.1(a) gives $\langle \lambda, \lambda \rangle - c_k = \langle \lambda, \lambda \rangle - 1$, i.e. $c_k = 1$. From the classification of minuscule posets, $c_k = \min(k, n + 1 - k)$ for $A_n\omega_k$, 2 for $D_n\omega_1$,

$\lfloor n/2 \rfloor$ for $D_n \omega_n$ ($n \geq 4$), 2 for $E_6 \omega_1$ and 3 for $E_7 \omega_7$; so $c_k = 1$ only for A_n with $k \in \{1, n\}$, whose poset is a chain. Conversely on a chain the ideals are the initial segments and R is the cyclic shift, a permutation matrix in $W(A_n) = S_{n+1}$; that this shift is the Coxeter element $s_1 \cdots s_n$ is classical (cf. [SW12, Thm 6.1], attributing the chain and product-of-chains cases to Stanley). $\square$

Theorem 3.3 (= Theorem A, second part). *Let h be even and $m = h/2$. Then $R^m \in W$ if and only if P is a chain or λ is the vector representation ω_1 of D_n , $n \geq 3$. In the vector case, on the weights $\pm e_i$,*

$$R^{n-1}(e_i) = -e_{n-i} \quad (1 \leq i \leq n-1), \quad R^{n-1}(e_n) = (-1)^{n-1} e_n,$$

a signed permutation with an even number of sign changes, hence in $W(D_n)$.

Proof. (Obstruction.) If $R^m \in W$ it is an isometry with $R^m(w(j)) = w(j+m)$, so Lemma 3.1(b) gives $\sigma_{j+m} = \sigma_j$ for all j : σ is m -periodic, and since $\sigma_0 = s_0 = 1$, necessarily $\sigma_m = 1$. For $A_n \omega_k$ with $1 < k < n$ (n odd, $m = (n+1)/2$), P is the $k \times (n+1-k)$ rectangle and $s_m = \min(k, n+1-k) \geq 2$. For $D_n \omega_n$, $n \geq 5$ ($m = n-1$), P is the shifted staircase $\{(i, j) : 1 \leq i \leq j \leq n-1\}$ with rank $i+j-2$, and $s_{n-1} = \lfloor (n+1)/2 \rfloor - 1 \geq 2$. For $E_6 \omega_1$ ($m = 6$) the rank sizes are 1, 1, 1, 2, 2, 2, 2, 2, 1, 1, 1 and $s_6 = 2$; for $E_7 \omega_7$ ($m = 9$) they are 1, 1, 1, 1, 2, 2, 2, 2, 3, 2, 2, 2, 2, 1, 1, 1, 1 and $s_9 = 2$. For $D_n \omega_1$ ($m = n-1$), P is a chain of $n-2$ elements above the antichain $\{e_n, -e_n\}$ above a chain of $n-2$, with $c_1 = 2$, and $\sigma = (1^{n-2}, 2, 1^{n-2}, 2)$ is $(n-1)$ -periodic; the triality images at $n = 4$ and the chains pass likewise.

(Sufficiency.) On $D_n \omega_1$ the ideals are $\emptyset$, the prefixes of the lower chain, the three ideals adding $e_n, -e_n$ or both, and the extensions along the upper chain; rowmotion cycles $\emptyset \rightarrow \cdots \rightarrow P$ in $2n-2$ steps and swaps the two one-sided ideals at every step. On weights,

$$-e_1 \rightarrow -e_2 \rightarrow \cdots \rightarrow -e_{n-1} \rightarrow e_{n-1} \rightarrow \cdots \rightarrow e_1 \rightarrow -e_1, \quad e_n \leftrightarrow -e_n.$$

The cycle has length $2n-2$ with $-e_i$ at position $i-1$ and e_j at position $2n-2-j$; shifting by $n-1$ sends $-e_i$ to position $n+i-2 = 2n-2-(n-i)$, i.e. to e_{n-i} . Hence $R^{n-1}(\pm e_i) = \mp e_{n-i}$ for $i \leq n-1$ and $R^{n-1}(e_n) = (-1)^{n-1} e_n$: the coordinate permutation $i \leftrightarrow n-i$ with $n-1$ sign changes on the first $n-1$ coordinates and one more iff n is even — n or $n-1$ sign changes, even either way. $\square$

Corollary 3.4. *$R^j \in W$ implies that σ is j -periodic; so the linear powers of rowmotion are among the periods of σ . For E_7 , σ has no period below 18: no power of R but the identity is a Weyl element.*

Theorem 3.5 (= Theorem B). *Let $\kappa(R)$ be the fraction of unordered pairs $\{w, w'\}$ of weights with $\langle Rw, Rw' \rangle = \langle w, w' \rangle$. For $n \geq 3$,*

$$\kappa(D_n \omega_1) = 1 - \frac{2(n-1)}{n(2n-1)}, \quad \kappa(A_n \omega_2) = 1 - \frac{3n^2 - 9n + 4}{\binom{n+1}{2}}.$$

Proof. ($D_n \omega_1$.) The inner products are 0 and -1 , the latter on the n antipodal pairs. By the cycle in the proof of Theorem 3.3, $\{e_n, -e_n\}$ is the only antipodal pair R keeps; the other $n-1$ become non-antipodal, and since R permutes the weights bijectively, $n-1$ non-antipodal pairs become antipodal: $2(n-1)$ of the $n(2n-1)$ pairs change.

($A_n \omega_2$.) Weights are the 2-subsets $\{a < b\}$ of $[n+1]$ and a pair's inner product is determined by $|S \cap T| \in \{0, 1\}$. The ideals $(x_1 \geq x_2)$ of the $2 \times (n-1)$ rectangle correspond to $\{n-x_1, n+1-x_2\}$, and rowmotion on the rectangle gives

$$R\{a, b\} = \{a-1, b-1\} \quad (a \geq 2, b \geq a+2), \quad R\{a, a+1\} = \{a-1, n+1\} \quad (a \geq 2),$$

$$R\{1, b\} = \{b-2, b-1\} \quad (b \geq 3), \quad R\{1, 2\} = \{n, n+1\}.$$

Let ρ be the rotation $i \mapsto i - 1 \pmod{n+1}$, a Coxeter element and an isometry. Then $R = \rho \circ \varphi$, where φ fixes every 2-subset outside $T = \{\{a, a+1\} : 2 \leq a \leq n\} \cup \{\{1, b\} : 2 \leq b \leq n+1\}$ and permutes T by $\{a, a+1\} \mapsto \{1, a\}$, $\{1, b\} \mapsto \{b-1, b\}$ ($b \geq 3$), $\{1, 2\} \mapsto \{1, n+1\}$; so $\kappa(R) = \kappa(\varphi)$. A pair $(S \in T, G \notin T)$ changes type iff $a+1 \in G$ (for $S = \{a, a+1\}$), iff $b-1 \in G$ (for $S = \{1, b\}$, $b \geq 3$), or iff exactly one of $2, n+1$ lies in G (for $S = \{1, 2\}$): $2(n-2)^2 + 2(n-3)$ pairs. Inside T , of the $(n-2) + 2(n-1) + \binom{n}{2}$ intersecting pairs exactly $4n-5$ remain intersecting, so $(n-1)(n-2)$ pairs change. The total is $3n^2 - 9n + 4$. $\square$

Every step of these proofs was re-checked on the data **(C)**: Lemma 3.1 on all 42 cases of Table 1; $c_k = 1 \Leftrightarrow \text{chain} \Leftrightarrow R \in W$; the periodicity criterion with exactly the predicted case list; the vector formula for $n = 3, \dots, 9$; the explicit rowmotion on 2-subsets and both formulas of Theorem 3.5 for $n = 3, \dots, 9$.

Remark 3.6 (observed). The analogous count for $A_n \omega_3$ is, for $n \leq 10$, the degree-4 polynomial $12D_3(n) = 15n^4 - 110n^3 + 165n^2 + 314n - 432$, suggesting degree $2k-2$ for ω_k . We have no proof.

3.1 The defect across the family

One program built from the Cartan matrix alone (weights in Dynkin labels, the lattice by $w \mapsto w - \alpha_i$ at label 1, its join-irreducibles, rowmotion as the join of the minimal elements of the complement) computes, for the 42 simply-laced minuscule cases with A_n ($2 \leq n \leq 7$, all k ; the trivial A_1 omitted), D_n ($n = 4, 5, 6, 7$, ω_1 and both half-spins), E_6 and E_7 : κ ; the random baseline $\kappa_0 = \sum_t p_t^2$ over the distribution p_t of inner-product types; the best pointwise agreement $a(R)$ of R with a Coxeter element; the *altitude* $v(R)$, the Hamming distance from R to the nearest element of W acting on the weights (W enumerated as permutations of the weights where $|W| \leq 322,560$; for E_7 by a base-and-strong-generating-set search over all 2,903,040 elements); and the linear centralizer $C_W(R)$.

Table 1. Rowmotion against the Weyl group on the simply-laced minuscule boards of rank ≤ 7 (dual representations give equal rows).

case	$ W\lambda $	h	cycle type of R	$\kappa(R)$	κ_0	$a(R)$	$v(R)$	$ C_W(R) $
$A_n \omega_1$	$n+1$	$n+1$	$(n+1)$	1	1	$n+1$	0	$n+1$
$A_3 \omega_2 =$ $D_3 \omega_1$	6	4	$4 \cdot 2$	11/15	0.680	3	3	2
$A_4 \omega_2$	10	5	5^2	29/45	0.556	5	5	1
$A_5 \omega_2$	15	6	$6^2 \cdot 3$	71/105	0.510	6	9	1
$A_5 \omega_3$	20	6	$6^3 \cdot 2$	58/95	0.452	7	13	1
$A_6 \omega_2$	21	7	7^3	76/105	0.500	10	11	1
$A_6 \omega_3$	35	7	7^5	339/595	0.419	10	25	1
$A_7 \omega_2$	28	8	$8^3 \cdot 4$	145/189	0.506	15	13	1
$A_7 \omega_3$	56	8	8^7	431/770	0.405	12	44	1
$A_7 \omega_4$	70	8	$8^8 \cdot 4 \cdot 2$	417/805	0.380	12	56	1
$D_4 \omega_1$	8	6	$6 \cdot 2$	11/14	0.755	4	3	2
$D_5 \omega_1$	10	8	$8 \cdot 2$	37/45	0.802	5	5	2
$D_5 \omega_5$	16	8	8^2	19/30	0.556	5	10	1
$D_6 \omega_1$	12	10	$10 \cdot 2$	28/33	0.835	6	5	2
$D_6 \omega_6$	32	10	$10^3 \cdot 2$	79/124	0.469	9	19	1
$D_7 \omega_1$	14	12	$12 \cdot 2$	79/91	0.858	7	7	2
$D_7 \omega_7$	64	12	$12^5 \cdot 4$	295/504	0.432	13	48	1
$E_6 \omega_1$	27	12	$12^2 \cdot 3$	223/351	0.527	9	15	1
$E_7 \omega_7$	56	18	$18^3 \cdot 2$	501/770	0.482	14	38	1

Rush–Shi’s conjugacy is visible in every row (the cycle type is that of a Coxeter element). κ exceeds κ_0 in every non-chain case. The linear centralizer is trivial except on chains and on the vector representations, where it is $\{1, R^{h/2}\}$ (Theorem 3.3). **Remark 3.7.** κ is not monotone in the rank along every family: it increases along $D_n \omega_1$

and $A_n\omega_2$ (Theorem 3.5) and decreases along the half-spins. E_6 and E_7 sit where the D_5 – D_7 spinors sit; the E_7 board is, by every measure in the table, as far from linear as rowmotion on a minuscule poset gets at this rank.

4. The E_7 board

Let $\Lambda = W\lambda$ be the 56 weights of the minuscule representation of E_7 , $W = W(E_7)$ of order $2,903,040 = 2 \times |Sp_6(2)|$, P the 27-element minuscule poset, and R rowmotion, of order 18 with cycle type $18^3 \cdot 2$. Since $-1 \in W(E_7)$, $\iota := -1 = w_0$ is central in W and pairs the weights into 28 antipodal pairs. With $\langle \alpha, \alpha \rangle = 2$ the weights have norm $3/2$ and two distinct weights have inner product $1/2$, $-1/2$ or $-3/2$, the last exactly for antipodal pairs; each weight has 27 partners of each non-antipodal type. Write $B(u, u') \in \mathbb{F}_2$ for the *type* of a non-antipodal pair, 0 if $\langle u, u' \rangle = 1/2$ and 1 if $-1/2$. B is the restriction to Λ of the symmetric bilinear form of the $\mathbb{F}_2$ -model of §8.3, in which the 56 weights are the nonsingular vectors u with $B(u, v) = 1$; in particular B is bilinear on the $\mathbb{F}_2$ -span of Λ , and $B(\beta, w) = \langle \alpha, w \rangle \bmod 2$ for a simple root α with image β .

4.1 The parity rule

For $u \in \Lambda$ write $\delta(u) := u + R(u)$ (in $\mathbb{F}_2$ -coordinates); $\delta(u)$ is the sum of the simple roots β_c , $c \in T(u)$, toggled at u . The toggle sizes on the 56 are $\{1 : 8, 2 : 28, 3 : 11, 4 : 2, 5 : 3, 6 : 2, 9 : 1, 27 : 1\}$.

Proposition 4.1 (= Theorem C). *For a non-antipodal pair $\{u, u'\}$, $B(Ru, Ru') = B(u, u')$ iff $t_1 + t_2 + t_3$ is even, where $t_1 = \#\{c \in T(u) : \text{extlabel}_c(u') \neq 0\}$, t_2 likewise with the roles exchanged, and t_3 is the number of Dynkin-adjacent pairs $(c, c') \in T(u) \times T(u')$.*

Proof. B is bilinear, so

$$B(Ru, Ru') - B(u, u') = B(\delta u, u') + B(u, \delta u') + B(\delta u, \delta u').$$

With $\delta u = \sum_{c \in T(u)} \beta_c$ we have $B(\beta_c, w') = \langle \alpha_c, w' \rangle \bmod 2 = |\text{extlabel}_c(w')|$ for minuscule labels in $\{-1, 0, 1\}$, and $B(\beta_c, \beta_{c'})$ is the Cartan entry mod 2, i.e. Dynkin adjacency. The three terms are t_1, t_2, t_3 . $\square$

The content of the proposition is therefore the *count*: on all 1,485 non-antipodal pairs the rule has zero exceptions (C), and with the 28 antipodal pairs the kept count is 1,002 of 1,540, i.e. $\kappa(R) = 501/770$ as in Table 1. The eight cells (t_1, t_2, t_3) and their populations are tabulated in the verification log; the dominant kept cell is $(1, 1, 0)$ with 553 pairs. The same rule with the k -fold toggle multiset is exact for every power R^k , $k = 1, \dots, 17$, and reproduces the measured profile $\kappa(R^9) = 0.616$, $\kappa(R^2) = 0.547$, $\kappa(R^3) = 0.513$ against a random mean of 0.482. Each 27-sheet (colour 7 deleted) is $J(P_6)$ with $|P_6| = 16$, its rowmotion R_6 has order 12 with orbits $12, 12, 3$, its pair types are the Schläfli adjacency, and the same rule with the E_6 diagram is exact there; $\kappa(R_6) = 223/351 = 0.635 < \kappa(R)$ — smaller toggle sets do not mean fewer flips.

4.2 What the Weyl group shares with the clock on the 56

Proposition 4.2 (C). $C_{S_{56}}(R) \cap W = \{1\}$: no non-trivial Weyl element commutes with R . More strongly, R agrees pointwise with the nearest Weyl element on 18 of the 56 weights, so $v(R) = 38$, and with the nearest Coxeter element on 14 (Table 1); the powers have altitudes $v(R^9) = 36$, $v(R^2) = v(R^6) = 44$, $v(R^3) = 46$.

Theorem 4.3 (= Theorem E(i)) (C).

$$W \cap R^k W R^{-k} = \{1\} \quad (k = 1, \dots, 8, 10, \dots, 17), \quad W \cap R^9 W R^{-9} = \{1, \iota\};$$

$$\langle W, R \rangle = A_{56}, \quad \langle W, \text{pr} \rangle = 2^{28} \rtimes Sp_6(2), \quad \langle W, R, \text{pr} \rangle = S_{56},$$

where pr is the involution of §4.3.

Proof (C). I_k was computed by conjugating every element of W by R^k and testing membership in W (a permutation of Λ is in W iff it extends to an isometry of the weight configuration; for E_7 this criterion is exact because the isometry group of Λ is W itself); for $k \neq 9$ the cheap necessary test — the conjugate must commute with the central ι — already leaves only the identity. That $\iota \in I_9$ is forced: R^9 commutes with ι (Theorem D) and ι is central in W . R , ι and all reflections are even permutations of the 56 and pr is odd; the group orders were established by Schreier–Sims. For $\langle W, \text{pr} \rangle$: pr fixes exactly the two poles, which form an antipodal pair, and $\text{pr} \circ t$ (t the pole swap) is a Weyl element (§4.3); W is transitive on the 28 antipodal pairs with image $Sp_6(2)$ and kernel $\langle \iota \rangle$, so the normal closure of t in $\langle W, \text{pr} \rangle$ is the group 2^{28} of all pair-flips, meeting W in $\langle \iota \rangle$. $\square$

Corollaries: no reflection of W (0 of 63) is a symmetry after any tick $k = 1, \dots, 17$, and none of the 21 involutions of a bridge subgroup $PSL(2, 7) \leq W$ (§9) is; the eighteen conjugates $R^k W R^{-k}$ are eighteen copies of $W(E_7)$ in S_{56} meeting pairwise in the identity (in $\langle \iota \rangle$ at distance 9).

4.3 The mirror

The coweight $\omega_7^\vee$ takes the values $\pm \frac{3}{2}$ on two weights (the *poles*) and $\pm \frac{1}{2}$ on two sets of 27 (the *sheets*); each sheet is the weight lattice of a 27-dimensional representation of E_6 , with its own rowmotion R_6 and its own longest element $w_0(E_6) \in W(E_6) < W(E_7)$, acting on the sheet's E_6 -labels as $-\sigma_6$, where σ_6 is the E_6 diagram automorphism. Define

$$\text{pr} := \iota \circ w_0(E_6) \circ t, \quad t = \text{the transposition of the two poles.}$$

Proposition 4.4 (C). (i) pr is an involution swapping the two sheets and fixing the poles; on the sheets it is the order isomorphism $J(P_6^{(0)}) \rightarrow J(P_6^{(1)})$ whose induced map on join-irreducibles is the colour map $0 \leftrightarrow 5, 2 \leftrightarrow 4, 1, 3$ fixed — the E_6 diagram automorphism (324 of 324 comparable pairs preserved each way, principal ideals to principal ideals). (ii) pr commutes with R_6 and with ι , and $\text{pr} \circ t = \iota \circ w_0(E_6)$ is a fixed-point-free involution of $W(E_7)$ at Hamming distance 2 from pr (so $v(\text{pr}) = 2$). (iii) pr preserves the inner-product type of exactly 1,432 of the 1,540 pairs — the count of an isometry composed with one transposition. (iv) $|\langle R, \iota \rangle| = 36$, $|\langle \text{pr}, \iota \rangle| = 4$, and $\langle R, \text{pr} \rangle = S_{56}$.

That the diagram automorphism swaps the two 27's and induces an isomorphism of their minuscule posets is standard [Gre13; GX26]; (ii) follows from Theorem D applied on each sheet ($w_0(E_6)$ reverses R_6 , ι reverses both R and R_6 , and a product of two reversers commutes). (iv) was established by Schreier–Sims; the Cayley diameter of $\{R, \text{pr}\}$ in S_{56} is at least 157 by counting.

Proposition 4.5 (the reversers on the 56) (C). Exactly one element of $W(E_7)$ reverses R , namely ι ; exactly one element of $W(E_6)$ reverses R_6 , namely $w_0(E_6)$. Hence pr is the product of the two reversals, corrected at the poles.

4.4 The bridge subgroup and the outer automorphism

Let $L \leq W(E_7)$ be a copy of $PSL(2, 7)$ transitive on the 28 antipodal pairs (§9 constructs it; its normaliser in $Sp_6(2)$ is computed in Proposition 9.1(iv)).

Proposition 4.6 (C). $C_{W(E_7)}(L) = \langle \iota \rangle$ and $N_{W(E_7)}(L)$ has order 672, so $N/C \cong PGL(2, 7)$: $W(E_7)$ realises every automorphism of L and no further symmetry of it. $N = \langle \iota \rangle \times PGL(2, 7)$, with one $PGL(2, 7)$ complement inside $Sp_6(2) = W^+$ (fixed-point-free on Λ) and one outside. The outer coset contains 56 involutions (28 of cycle type $2^{24}1^8$, 28 fixed-point-free), 112 elements of order 6 and 168 of order 8. Any $c \in N \setminus L(\iota)$ is a Weyl element and therefore commutes with no power of R and reverses none (Proposition 4.2, Theorem 4.3): the unique reverser of R in N is ι and the unique element of N commuting with R is 1.

P acts on the 28 antipodal pairs with permutation character $\chi_1 + 2\chi_6 + \chi_7 + \chi_8$ (fixed points 28, 4, 1, 0, 0 on the classes 1A, 2A, 3A, 4A, 7AB), and $\mathbb{C}\Lambda = 2(\chi_1 + 2\chi_6 + \chi_7 + \chi_8)$ as a P -module; the three-dimensional irreducibles $\chi_3, \bar{\chi}_3$ do not occur.

5. The shared grammar across the family

5.1 Reversal

Theorem 5.1 (= Theorem D). *On every simply-laced minuscule board the longest element w_0 , acting on weights as $-\sigma$ (σ the diagram automorphism carrying λ to λ^* ; trivial when $-1 \in W$), reverses rowmotion:*

$$w_0 R w_0 = R^{-1}.$$

Proof. The antipode $w \mapsto -w$ reverses the root order on $W\lambda \cup W\lambda^*$ and σ preserves it, so $w_0 = -\sigma$ is an anti-automorphism of the weight lattice, hence of $J(P)$. Rowmotion sends an ideal I to the ideal generated by the minimal elements of $P \setminus I$; its inverse sends I to the ideal whose complement is the filter generated by the maximal elements of I . An anti-automorphism of $J(P)$ exchanges ideals with filters and minimal with maximal elements, so it exchanges the two recipes. $\square$

Checked on all 42 boards **(C)**: $w_0 \in W$, $w_0(\lambda) = w_0\lambda$, $w_0 R w_0 = R^{-1}$.

Corollary 5.2. (i) *The Weyl elements reversing R form the coset $w_0 C_W(R)$ (if g, g' both reverse R then $g^{-1}g'$ commutes with it); on the rich boards the reverser is unique. (ii) w_0 commutes with R^k iff $R^{2k} = 1$; for h even, $w_0 \in I_{h/2}$ always. (iii) *On a self-dual board the antipode ι commutes with R^k iff $R^{2k} = 1$, and R^k preserves the pair $\{w, -w\}$ iff $R^{2k}w = w$; so the antipodal pairs preserved at lag k number $\frac{1}{2}\text{fix}(R^{2k})$, and R^k induces a map on $W\lambda/\pm$ only when $R^{2k} = 1$. On E_7 : one pair for $k = 1, \dots, 8$ and all 28 at $k = 9$.**

Proposition 5.3 (what the reverser does to W) (C). $w_0 s_i w_0 = s_{\sigma(i)}$ on every board. The reverser is central in W , and equal to the antipode, on exactly seven boards of rank ≤ 7 — $D_4 \omega_1/\omega_3/\omega_4$, $D_6 \omega_1/\omega_5/\omega_6$, $E_7 \omega_7$ — and on every other board it is non-central and acts on W by the diagram flip ($\Lambda^k \leftrightarrow \Lambda^{n+1-k}$ on A_n , the two half-spins on D_{odd} , $27 \leftrightarrow \overline{27}$ on E_6). On the five self-dual boards with $-1 \notin W$ ($A_3\omega_2$, $A_5\omega_3$, $A_7\omega_4$, $D_5\omega_1$, $D_7\omega_1$) the antipode is a permutation of the weights that reverses R and is not a Weyl element, and the isometry group of the weight configuration is $W \times \langle -1 \rangle$; on all other boards it is W .

5.2 The centralizer law and its exceptions

Proposition 5.4 (= Theorem E(iii)) (C). *For the 42 boards of Table 1 with W enumerated in full: on every board on which no power of R is a Weyl element — the middle representations ω_k ($1 < k < n$) of $A_4, \dots, A_7$, the half-spins of D_5 , D_6 and D_7 , E_6 and E_7 : 23 boards — $I_k = C_W(R^k)$ as sets for every k , with one exception, the two half-spins of D_5 , where $|I_3| = |I_5| = 2$ against $|C_3| = |C_5| = 1$ and $|I_4| = 8$ against $|C_4| = 4$. On the chains $I_k = W$. On the vector boards $D_n\omega_1$, $|I_k|$ follows $\gcd(k, n-1)$ (on D_7 : 12, 72, 384, $|W|$ for $\gcd = 1, 2, 3, 6$, against centralizers 2, 12, 16, 384).*

The chains and the vectors are explained by Theorems 3.2–3.3 and by §5.3; the D_5 exception is Theorem F (§6).

Remark 5.5. The statement “ $I_k = C_k$ on every board” is false; the statement “the unique non-trivial half-turn survivor on a rich board with h even is -1 when $-1 \in W$ and nothing otherwise” is also false — the survivor is w_0 (Corollary 5.2(ii)), which is -1 only when $-1 \in W$. On the D_6 half-spins four elements survive the half-turn ($w_0 = -1$ among them) and on the D_5 half-spin eight.

5.3 The vector boards: the centralizer of the doubled clock

Lemma 5.6. *If an involution ι of the weights centralizes W and reverses R , then $I_k \subseteq C_W(R^{2k})$ for every k .*

Proof. For $g \in I_k$ let $h = R^{-k}gR^k \in W$. Then $\iota h \iota = R^k g R^{-k}$ (since $\iota R \iota = R^{-1}$ and $\iota g \iota = g$), while $\iota h \iota = h$ because $h \in W$ and ι centralizes W . Hence $R^k g R^{-k} = R^{-k} g R^k$, i.e. g commutes with R^{2k} . $\square$

The hypothesis holds on the seven boards with $-1 \in W$ and on the five self-dual boards of Proposition 5.3 whose antipode is not a Weyl element (there $\iota = -1$ centralizes W as a linear map); the inclusion was checked at every lag, with W enumerated, on the eleven of these boards other than E_7 and on the B_4 spinor **(C)**; on E_7 it follows from Theorem 4.3.

Theorem 5.7 (= Theorem E(ii)). On the vector boards $D_n \omega_1$ ($n \geq 3$, weights $\pm e_i$), $I_k = C_W(R^{2k})$ for every k .

Proof. $\subseteq$ is Lemma 5.6 with $\iota = -1$ (a Weyl element for n even, the antipode otherwise; in both cases it centralizes W and reverses R by Theorem 5.1 and Proposition 5.3). For $\supseteq$ let $g \in W$ commute with R^{2k} and put $h = R^{-k}gR^k$. Then $\iota h = R^k g R^{-k} = R^{-k}(R^{2k}gR^{-2k})R^k = h$, so h is a permutation of $\{\pm e_i\}$ commuting with -1 , i.e. a signed permutation, an element of $W(B_n)$ (the centralizer of -1 in $\text{Sym}(2n)$). Since h is conjugate to g in $\text{Sym}(2n)$ it has the same sign, and the sign character of $\text{Sym}(2n)$ restricted to $W(B_n)$ is $(-1)^{\#\text{negative cycles}}$, whose kernel is $W(D_n)$; so $h \in W(D_n) = W$ and $g \in I_k$. $\square$

Proposition 5.8 (the transports on the vector boards) (C, with proof of the first sentence). A transport $g \mapsto R^{-k}gR^k$ preserves the unsigned cycle type and the sign of a signed permutation, so it can change the signed cycle type only by trading two negative j -cycles for one positive $2j$ -cycle; no element of $W(B_n)$ realises such a trade by conjugation, and g is W -inner (its transport equals $w^{-1}gw$ for some $w \in W$) iff the signed cycle types of g and its transport agree. At every lag coprime to $n - 1$ the shared grammar I_k is dihedral of order $2(n - 1)$ and the transport preserves its cyclic half; for n odd it is the outer automorphism swapping the two reflection classes (inner in no group below $\text{Sym}(2n)$), and for n even it is conjugation by $R^{h/2}$ or by $w_0 R^{h/2}$ — because on $C_W(R^{2k})$ the transport equals conjugation by any odd power of R^k , and $h/2 = n - 1$ is such a power exactly when n is even. The non-inner counts: D_3 , 2 of 4; D_5 , 4 of 8 at odd lags and 18 of 32 at lags 2, 6; D_7 , 6 of 12 at lags coprime to 6 and 252 of 384 at lags 3, 9; none on D_4 , D_6 .

Remark 5.9 (a false lemma). The identity $I_k = \bigcup_{w \in W} C_W(R^k w^{-1})$ is tempting and false. The inclusion $\supseteq$ holds; equality holds iff every transport is W -conjugate to its element. It fails on the B_3 spinor (8 of the 16 elements of I_3), on $A_3 \omega_2$ and on the odd D_n vectors — exactly the non-inner transports of Proposition 5.8. Across the 98 (board, lag) pairs of the 42 boards on which $I_k \neq C_W(R^k)$, on 83 a single Weyl element realises the transport and on 15 some transports leave their W -class.

6. The 4-cube

6.1 Locating the exception

The D_5 half-spin (16 weights, $h = 8$, R two free 8-cycles) shares its poset and its rowmotion with the B_4 spinor board — the 16 vertices of the 4-cube under $W(B_4) = \mathbb{F}_2^4 \rtimes S_4$ of order 384 — and $W(B_4) < W(D_5)$ (C). The half-turn anomaly is already present on B_4 : $|I_4| = 8$ against $|C_4| = 4$, with the same four extra elements — two signed 4-cycles $e_1 \rightarrow -e_4 \rightarrow -e_2 \rightarrow e_3 \rightarrow e_1$ and its inverse, and two involutions pairing (e_1, e_4) with (e_2, e_3) . Only the lag-3 survivor needs the fifth axis: it is the coordinate permutation $(e_1 e_3)(e_2 e_5)$, the one survivor that mixes the clock's two orbits. What the anomaly is *not* (C): not a hidden $\mathbb{F}_2$ -affinity (no power of the D_5 , D_6 or D_7 clock is an affine map of the even code; D_4 's half-turn is, being the linear half-turn of Theorem 3.3); not a near-miss of linearity (the altitude of R^4 against $W(D_5)$ is 8 of 16, of R^3 7); not an overgroup effect ($\langle W(D_5), R^4 \rangle = A_{16}$; contrast the B_3 spinor, whose half-turn signature — shared grammar 16 against centralizer 8 — comes from $W(D_4) \supset W(B_3)$ containing the D_4 half-turn); and not specific to the 4-cube in the naive sense (the B_5 spinor is clean, $I_k = C_k$ for every k).

6.2 The mechanism

Lemma 6.1. Let $w \in W$ and suppose $Y := \text{Fix}(R^k w^{-1})$ has trivial pointwise stabilizer in W . Then $I_k \cap \text{Stab}_W(Y) = C_W(R^k w^{-1})$, and on this set the transport is conjugation by w .

Proof. Put $c = R^k w^{-1}$, so $R^k = cw$ and c is the identity on Y . If $g \in W$ commutes with c then $R^{-k}gR^k = w^{-1}c^{-1}gcw = w^{-1}gw \in W$, so $g \in I_k$, and g preserves $Y = \text{Fix}(c)$. Conversely let $g \in I_k$ preserve Y and let $h = R^{-k}gR^k \in W$; then $c^{-1}gc = whw^{-1} \in W$. For $y \in Y$, $c(y) = y$ and $g(y) \in Y$, so $c^{-1}gc(y) = c^{-1}g(y) = g(y)$; the two Weyl elements $c^{-1}gc$ and g agree on Y , hence are equal by the hypothesis on Y , i.e. g commutes with c ; and then $h = w^{-1}gw$. $\square$

Theorem 6.2 (= Theorem F). On the B_4 spinor board, R^4 agrees with $-\tau$, $\tau = (e_2 e_3)(-e_4)$, on all of O_2 and with no Weyl element on $O_1 \ni \lambda$. Consequently $I_4 = C_W(c)$ for the correction $c = (-\tau)R^4$, an involution that is the

identity on O_2 and fixed-point-free on O_1 ; $|I_4| = 8$, the four extras being the elements commuting with c but not with τ ; the transport on I_4 is conjugation by τ ; and $[g, R^4] = -1$ for each extra g .

Proof. Grade the weights by rank = wt(x). For a Weyl element $w = (\pi, v)$ put $u := \pi^{-1}(v)$; since a coordinate permutation preserves Hamming weight,

$$\text{rank}(w(x)) = \text{wt}(\pi(x) \oplus v) = \text{wt}(x \oplus u), \quad \Delta_w(x) := \text{rank}(w(x)) - \text{rank}(x) = \text{wt}(u) - 2|x \cap u|.$$

The orbit O_1 of $\lambda = 0$ has ranks 0, 4, 3, 3, 2, 2, 1, 1 in the order $\lambda, R\lambda, R^2\lambda, \dots$ (the *full-height* orbit: R sends λ straight to $w_0\lambda$ by Lemma 3.1(a), and the ranks then descend in pairs), and R^4 pairs clock positions j and $j + 4$, whose ranks differ by exactly 2. If $R^4|_{O_1} = w|_{O_1}$ then $\Delta_w(x) = \pm 2$ on O_1 ; at $x = 0$, which R^4 moves up to rank 2, $\Delta_w(0) = \text{wt}(u) = 2$; then $\Delta_w(x) = 2 - 2|x \cap u| \in \{2, 0, -2\}$ and $|\Delta_w| = 2$ forces $|x \cap u| \neq 1$, i.e. $\langle x, u \rangle = 0$, for every $x \in O_1$. But O_1 contains 0 and spans $\mathbb{F}_2^4$, so $u = 0$, contradicting $\text{wt}(u) = 2$. On O_2 the ranks stay in $\{1, 2, 3\}$ and R^4 shifts each by ± 1 ; $u = 1110$ gives $\Delta = 3 - 2|x \cap u| \in \{1, -1\}$ for all $x \in O_2$, and the Weyl element with this u and the matching translation is $-\tau$, which therefore agrees with R^4 on O_2 (uniquely, O_2 spanning). O_2 affinely spans $\mathbb{F}_2^4$, so its pointwise stabilizer in W is trivial, and Lemma 6.1 with $Y = O_2$, $w = -\tau$ gives $I_4 \cap \text{Stab}_W(O_2) = C_W(c)$. Finally the sixteen Weyl elements that swap the two orbits (of the 384, sixteen stabilise each orbit, sixteen swap them, and the rest mix them) lie in no I_k (checked **(C)**; by hand, $\text{Stab}_W(O_1)$ is sixteen affine maps whose image in S_4 is the dihedral stabilizer of the pairing $\{14|23\}$, kernel ± 1 , and I_4 is the preimage of its Klein four-group), so $I_4 = C_W(c)$, of order 8. The remaining assertions are Lemma 6.1's transport statement and direct computation. $\square$

The lag-3 survivor of D_5 falls to the same lemma with the nearest Weyl element w_3 (unique at Hamming distance 7; its agreement set of nine points determines $W(D_5)$): $(e_1 e_3)(e_2 e_5)$ commutes with $R^3 w_3^{-1}$, and its lag-5 partner is its transport **(C)**. The anomaly sits on B_4 , where the reverser -1 is central, so it is not tied to D_5 's non-central reverser.

Lemma 6.3 (dimension-free form). *On the B_n spinor board, if an R -orbit O contains $\lambda = 0$, spans $\mathbb{F}_2^n$, and $R^{h/2}$ shifts every rank on O by a constant magnitude $m \geq 2$, then $R^{h/2}|_O$ is not the restriction of a Weyl element. (Proof. If $w = (\pi, v)$ agreed with $R^{h/2}$ on O , then at $\lambda = 0$ the shift is $\text{wt}(u) = m$, and $|\Delta_w(x)| = m$ forces $|x \cap u| \in \{0, m\}$: every $x \in O$ misses u or contains it. These two parallel affine subspaces have codimension m , so their union spans a subspace of dimension $n - m + 1 < n$, contradicting that O spans $\mathbb{F}_2^n$. $\square$) On the full-height orbit of B_n the half-turn shifts rank by exactly $\pm n/2$ for every even n (Proposition 6.13); the clock acts freely on the 2^n weights with orbits of size $2n$ only when $2n \mid 2^n$, i.e. n a power of 2, so the clean two-orbit dichotomy of B_4 next arises, if at all, on B_8 (256 weights, 16 orbits of 16). It does not (Proposition 6.7, §6.4).*

6.3 Iterated shared grammars

For a board and a lag k define $P_1(k) = W$ and $P_{j+1}(k) = \{g \in P_j : R^{-k} g R^k \in P_j\}$, so $P_2 = I_k$; let $d(k)$ be the first j with $P_{j+1} = P_j$ and $P_\infty(k)$ the stable value. (The definition mimics the Clifford hierarchy of quantum computation, with W in the role of the Clifford group; we use it only as a chain of subgroups.)

Lemma 6.4. $P_j(k) = \bigcap_{0 \leq i < j} R^{ik} W R^{-ik}$. Hence $P_\infty(k)$ is the largest subgroup of W normalised by R^k , and it contains the core of W in $\langle W, R^k \rangle$.

Lemma 6.5. $P_3(k) = P_2(k)$ iff $I_k = I_{-k}$ iff some (equivalently every) reverser of R in W normalises I_k .

Proof. $P_3 = W \cap R^k W R^{-k} \cap R^{2k} W R^{-2k}$, so $P_3 = P_2$ iff $I_k \subseteq R^{2k} W R^{-2k}$, iff (conjugating by R^{-k}) $I_{-k} \subseteq I_k$, iff $I_{-k} = I_k$ since the two have equal order. For a reverser r , $r I_k r = W \cap r R^k W R^{-k} r = W \cap R^{-k} W R^k = I_{-k}$. $\square$

Proposition 6.6 (C, 44 boards, 278 (board, lag) pairs). $d(k) = 1$ exactly where $I_k = W$ (the chains; the D_n vectors at the half-turn). $d(k) = 2$ everywhere else — two instants already share what all instants share — with exactly one exception, the D_5 half-spin at lags 3 and 5, where the survivor is carried to a different element, no reverser normalises I_3 , and the chain drops once more: $1920 \supset 2 \supset 1$. The stable groups: $C_W(R^k)$ on the rich boards (trivial off the half-turn; at the half-turn $\{\pm 1\}$ where $-1 \in W$, except on the D_6 half-spins, where it has order 4, Remark 5.5); $C_W(R^{2k})$ on the D_n vectors; the dihedral I_4 of order 8 on the 4-cube and the D_5 half-spin at

the half-turn; on E_7 , trivial at every lag but the half-turn, where it is $\{\pm 1\}$. Wherever P_∞ is neither W nor of order ≤ 2 (44 cases) it is not normal in W ; where $\langle W, R \rangle$ could be enumerated ($D_3\omega_1, D_4\omega_1$, the B_3 spinor: A_6, A_8, A_8) the core of W in $\langle W, R \rangle$ is trivial, below $P_\infty(1)$ (of orders 4, 6 and 1; equality in the last case).

6.4 The 8-cube, the 16-cube, and the commutation theorem

The clock acts freely on the 2^n weights of the B_n spinor board, with orbits of size $h = 2n$, only when $2n \mid 2^n$, i.e. when n is a power of 2; B_8 is therefore the next board on which the two-orbit mechanism of Theorem 6.2 could recur, and the last one within reach (B_{16} has 65,536 weights). $W(B_8) = \mathbb{F}_2^8 \rtimes S_8$ and $W(D_9)$ were enumerated in the affine form $x \mapsto \pi(x) \oplus v$ on the 256 sign patterns (for D_9 , the even code with its parity bit dropped), never as permutation tuples; membership of a transport $R^{-k}gR^k$ in W is decided by its values at 0 and the unit vectors and then confirmed on all 256 points. *Notation.* For a product π of disjoint transpositions of the coordinates $e_2, \dots, e_{n-1}$ we write $-\pi$ for the signed permutation $x \mapsto \pi(x) \oplus (e_1 + \dots + e_{n-1})$: it negates $e_1, \dots, e_{n-1}$, permutes them by π , and fixes e_n ; on B_4 , $-(e_2e_3)$ is the element $-\tau$ of Theorem 6.2. The correction is written $c = w^{-1}R^{h/2}$ rather than Lemma 6.1's $R^{h/2}w^{-1}$; since $\text{Fix}(R^{h/2}w^{-1}) = w \text{Fix}(w^{-1}R^{h/2})$ and in every case below w preserves its agreement set $Y = \text{Fix}(c)$ (the stabilizers computed are $\langle -1, w \rangle$, or w commutes with $R^{h/2}$), the two fixed sets coincide, Lemma 6.1 applies to Y verbatim, and $C_W(R^{h/2}w^{-1}) = w C_W(c) w^{-1}$ has the order stated.

Proposition 6.7 (C). *On the B_8 spinor board (256 weights, $|W| = 2^8 \cdot 8! = 10,321,920$, $h = 16$):*

(i) *R has sixteen free orbits of sixteen. The orbit $O_1 \ni \lambda$ is the full-height orbit, with ranks $0, 8, 7, 7, \dots, 1, 1$ in the order $\lambda, R\lambda, R^2\lambda, \dots$, and R^8 shifts every rank on it by exactly ± 4 ; by Lemma 6.3 no Weyl element agrees with R^8 on O_1 .*

(ii) *Exactly two orbits carry the half-turn as a Weyl element, both by the same element $w = -(e_2e_3)(e_4e_5)(e_6e_7)$, which negates $e_1, \dots, e_7$ and fixes e_8 — the analogue of $-\tau$. It is the unique Weyl element nearest to R^8 , agreeing with it on 52 of the 256 weights. Each carrying orbit is preserved by -1 . Every orbit affinely spans $\mathbb{F}_2^8$.*

(iii) *Lemma 6.1 applies with $Y = \text{Fix}(w^{-1}R^8)$ (52 points, spanning), but the correction $c = w^{-1}R^8$ is not an involution (cycle type $8^8 4^{12} 3^{12} 2^{28} 1^{52}$) and $C_W(c) = \{\pm 1\}$. Hence $I_8 = C_8 = \{\pm 1\}$, and $I_k = C_k$ for every $k = 1, \dots, 15$. The same holds at every lag on the D_9 half-spin (the same clock; $|W(D_9)| = 92,897,280$) and on the B_6 and B_7 spinor boards.*

Consequently, among the spinor boards B_n with $3 \leq n \leq 8$, the shared grammar exceeds the centralizer only on B_4 and, through the overgroup $W(D_4)$, on B_3 (§6.1). The mechanism of Theorem 6.2 — the half-turn agreeing with a Weyl element on an orbit that avoids the extreme ranks — recurs on B_8 ; its consequence does not, because the correction's centralizer collapses to $\{\pm 1\}$.

Remark 6.8. On B_8 the two carrying orbits are exactly the orbits whose ranks stay in $\{3, 4, 5\}$, equivalently those on which every half-turn rank-shift is ± 1 ; on B_4 the carrying orbit O_2 likewise has ranks $\{1, 2, 3\}$. On B_{16} the sixteen carrying orbits are again exactly the band $\{7, 8, 9\}$ (Proposition 6.11). This is an observation on three boards; the orbit of λ itself is described exactly in Proposition 6.13. The expectations that the phenomenon of Theorem 6.2 recurs on B_8 , that it marks the boards with a free clock ("n a power of 2"), that -1 exchanges the two carrying orbits, and that D_9 carries a survivor at a lag coprime to 16 were all inverted by the computation.

Lemma 6.9 (the half-turn criterion). *Let $m = h/2$, $w \in W$ and $c = w^{-1}R^m$. (i) $c^2 = 1$ if and only if $R^m w R^m = w^{-1}$; for an involution w this says $w \in C_W(R^m)$. (ii) If $w \in C_W(R^m)$ then w, c and R^m pairwise commute; for $g \in W$ any two of " g commutes with w ", " g with R^m ", " g with c " imply the third; and when $\text{Fix}(c)$ has trivial pointwise stabilizer, $I_m \cap \text{Stab}_W(\text{Fix } c) = C_W(c)$ with $(I_m \cap \text{Stab}_W(\text{Fix } c)) \setminus C_W(R^m) = C_W(c) \setminus C_W(w)$.*

Proof. (i) $c^2 = w^{-1}R^m w^{-1}R^m = 1 \iff R^m w^{-1}R^m = w \iff R^m w R^m = w^{-1}$, since R^m is an involution. (ii) $cw = w^{-1}R^m w = w^{-1}w R^m = R^m = wc$, and both commute with $R^m = wc$; if g commutes with two of w, c, wc it commutes with the third. The rest is Lemma 6.1 together with the observation that for $g \in C_W(c)$, $g \notin C_W(R^m)$ if and only if $g \notin C_W(w)$. $\square$

Proposition 6.10 (C). *On the spinor boards $B_3, \dots, B_8$, $I_{h/2} \supsetneq C_{h/2}$ if and only if a Weyl element nearest to $R^{h/2}$ (one maximising the agreement set $\text{Fix}(w^{-1}R^{h/2})$) commutes with $R^{h/2}$. On B_4 the nearest element is $-\tau$, it commutes with R^4 , c is an involution and the four extras are $C_W(c) \setminus C_W(-\tau)$. On $B_5, \dots, B_8$ the nearest element is unique,*

of the shape $-(e_2e_3)(e_4e_5) \cdots$ (with e_n fixed for n even, and e_{n-1}, e_n negated for n odd), does not commute with $R^{h/2}$, c is not an involution, and $C_W(c) = \{\pm 1\}$; on B_8 the stabilizer of the 52-point agreement set is $\langle -1, w \rangle$. On B_3 the four nearest elements all commute with R^3 and $|I_3| = 16 > |C_3| = 8$, so the equivalence holds there as well; but no element of $W(B_3)$ agrees with R^3 on an affinely spanning set, Lemma 6.1 does not apply, and the excess is explained instead by the overgroup $W(D_4) \ni R^3$ (§6.1).

Proposition 6.11 (C, the 16-cube at the orbit level). *On the B_{16} spinor board (65,536 weights, $h = 32$; W is not enumerated): R has 2,048 free orbits of 32; the orbit of λ is the full-height orbit and R^{16} shifts every rank on it by exactly ± 8 ; exactly sixteen orbits carry the half-turn as a Weyl element, all by the same $w = -(e_2e_3)(e_4e_5) \cdots (e_{14}e_{15})$ with e_{16} fixed, and they are exactly the orbits whose ranks lie in $\{7, 8, 9\}$; w does not commute with R^{16} ; $C_W(R^{16}) = \{\pm 1\}$; the agreement set $Y = \text{Fix}(w^{-1}R^{16})$ has 1,900 points and spans, $\text{Stab}_W(Y) = \langle -1, w \rangle$ and $C_W(w^{-1}R^{16}) = \{\pm 1\}$, so by Lemma 6.1 $I_{16} \cap \text{Stab}_W(Y) = \{\pm 1\}$. Whether $I_{16} = \{\pm 1\}$ is not decided. (Rowmotion was computed combinatorially on the shifted staircase and validated against the B_4 and B_8 clocks; the Weyl element agreeing with R^{16} on an orbit was found by matching columns of the difference vectors, the stabilizer and the centralizers by backtracking over coordinate permutations with every solution verified on all 65,536 weights. 1,987 of the 2,048 orbits affinely span $\mathbb{F}_2^{16}$, every carrying orbit among them.)*

Lemma 6.12 (rowmotion in excess coordinates). *Write a weight of the B_n spinor board by its plus positions $p_1 < \cdots < p_r$ and put $d_k = p_k - k$, so $0 \leq d_1 \leq \cdots \leq d_r \leq n - r$ (the empty sequence is $w_0\lambda$). Call k a gap row if $d_k > d_{k-1}$ ($d_0 := 0$). Then the excess sequence of $R(x)$ is*

$$d'_{k'} = \min \left(\{d_k - 1 : k \text{ a gap row, } k \geq k'\} \cup \{n - r - 1 : \text{if } d_r < n - r \text{ and } k' \leq r + 1\} \right),$$

for every k' for which the set is non-empty; rows with an empty set are absent.

Proof. Identify P_λ with the shifted staircase δ_n of §2: cells (k, c) with $1 \leq k \leq c \leq n$ and colour c , (k, c) covering $(k, c + 1)$ and $(k - 1, c - 1)$ — the k -th plus sign, counted from the left, having reached position c — so that the ideal of the weight with plus positions $p_1 < \cdots < p_r$ is $\{(k, c) : k \leq r, c \geq p_k\}$, a cell $(k, c) \notin I$ with $k \leq r$ has $c < p_k$ and is minimal in $P \setminus I$ iff its lower covers lie in I , i.e. iff $c = p_k - 1$ and $(k = 1 \text{ or } p_k - 1 \geq p_{k-1} + 1)$, which is $d_k > d_{k-1}$; the only candidate in row $r + 1$ is $(r + 1, n)$, minimal iff $r = 0$ or $p_r \leq n - 1$, i.e. $d_r < n - r$; rows $\geq r + 2$ contain no minimal element. The down-closure of (k, c) is $\{(k', c') : k' \leq k, c' \geq c - k + k'\}$, so the threshold of $R(I)$ in row k' is the minimum of $c - k + k'$ over the generators (k, c) with $k \geq k'$; subtracting k' gives the formula. $\square$

Proposition 6.13 (the orbit of λ). *For $1 \leq j \leq \lfloor n/2 \rfloor$, $R^{2j}\lambda = [n] \setminus \{n, n - 2, \dots, n - 2j + 2\}$, and for $0 \leq j \leq \lfloor (n - 1)/2 \rfloor$, $R^{2j+1}\lambda = [n] \setminus \{n - 1, n - 3, \dots, n - 2j + 1\}$ (weights written as sets of minus positions). Hence the ranks around the orbit of λ read $0, n, n - 1, n - 1, \dots, 1, 1$, and for even n the half-turn R^n shifts every rank on this orbit by exactly $\pm n/2$.*

Proof. In excess coordinates $R^{2j}\lambda$ is $d_k = n - 2j + k$ ($k \leq j$) and $R^{2j+1}\lambda$ is $d_k = n - 1 - 2j + k$; both are strictly increasing, so every row is a gap row, and one checks from Lemma 6.12 that the first has no cap and loses 1 in every entry, while the second has the cap $n - j - 1$, which is larger than every $d_k - 1$ and is appended as a new row. The base $R\lambda = w_0\lambda$ is the empty sequence. Ranks follow by counting, and the second half of the orbit from $R^{-k}\lambda = -R^{k+1}\lambda$ (Theorem 5.1 with $w_0\lambda = R\lambda$, Lemma 3.1). $\square$

Theorem 6.14 (why four; = Theorem G). *Let $n \geq 4$ be even and $w_n = -(e_2e_3)(e_4e_5) \cdots (e_{n-2}e_{n-1})$, the signed permutation negating $e_1, \dots, e_{n-1}$ and fixing e_n . Then w_n commutes with the half-turn R^n if and only if $n = 4$.*

Proof. Write O_1 for the orbit of λ . (a) w_n preserves O_1 : by Proposition 6.13 and the identity $R^{-k}\lambda = -R^{k+1}\lambda$, a direct computation gives $w_n(R^{2j}\lambda) = R^{2-2j}\lambda$ and $w_n(R^{2j+1}\lambda) = R^{-(2j+1)}\lambda$; the maps $i \mapsto 2 - i$ and $i \mapsto -i$ on $\mathbb{Z}/2n$ commute with $i \mapsto i + n$ when n is even, so $w_n R^n = R^n w_n$ on O_1 . (b) If R^n agrees with the involution w_n on an orbit O , then $w_n R^n x = x = R^n w_n x$ for $x \in O$. (c) For $n = 4$ the board is $O_1 \cup O_2$ and $R^4 = -\tau = w_4$ on O_2 (Theorem 6.2), so $w_4 \in C_W(R^4)$. (d) For even $n \geq 6$ let $e_2 = \{2\}$. Iterating Lemma 6.12 along the orbit of e_2 gives, for $2 \leq j \leq n/2 - 1$, the excess sequences $(n - 1 - 2j, n - 2j + 2, \dots, n - j)$ at $R^{2j}e_2$ and $(n - 2 - 2j, n - 2j + 1, \dots, n - j - 1)$ at $R^{2j+1}e_2$, and then $R^n(e_2) = \{1, 2\} \cup \{5, 7, \dots, n - 1\}$; along the orbit of $\{3, n\}$ (whose first step is e_4) it gives $R^n(\{3, n\}) = [n] \setminus \{4, 6, 7, 9, 11, \dots, n - 1\}$ for $n \geq 8$ and $\{1, 2, 3, 5, 6\}$ for $n = 6$. Since $\pi = (2\ 3)(4\ 5) \cdots$ swaps 2 and 3 and $3 \notin R^n(e_2)$, we get $2 \in w_n(R^n e_2) = \pi(R^n e_2) \triangle [n - 1]$; and

$w_n(e_2) = [n - 1] \setminus \{3\} = -\{3, n\}$, so $R^n(w_n e_2) = -R^n(\{3, n\}) \not\equiv 2$ because $2 \in R^n(\{3, n\})$. Hence $w_n R^n e_2 \neq R^n w_n e_2$. $\square$

The theorem is checked on the actual clocks for even $n \leq 16$, Lemma 6.12 on every weight for $n \leq 12$, and every excess sequence named in the proof for $6 \leq n \leq 16$ (C). Together with Lemma 6.9 it is the reason for the 4-cube: the mechanism of Theorem 6.2 is present on every free clock, and its consequence requires a commutation that only the smallest board allows, because there the two kinds of orbit on which commutation is forced exhaust the board.

7. Homomesy of the label counts

A statistic on weights is *homomesic* under R if its average is the same on every R -orbit [PR15]. Rush and Wang proved that on every minuscule poset the ideal cardinality and the antichain cardinality are homomesic under rowmotion, and that the per-colour ideal cardinalities are [RW15]; on the weights the latter says every orbit has mean weight zero.

A weight's label at node i is $+1$ iff its ideal has a maximal element of colour i , -1 iff the complement has a minimal element of colour i , and 0 otherwise. Hence $\#(+1) = |\max I|$, the antichain cardinality; $\#(-1) = |\min(P \setminus I)| = |\max R(I)|$; and $\#(0) = r - \#(+1) - \#(-1)$, r the rank.

Proposition 7.1. *On every simply-laced minuscule board the three ternary label counts are homomesic under R , with orbit means*

$$\#(+1) = \#(-1) = \frac{|P|}{h}, \quad \#(0) = r - \frac{2|P|}{h}.$$

The toggle size $|I \triangle R(I)|$ is homomesic with mean $2|P|/h$.

Proof. The first two counts are the antichain cardinality of I and of $R(I)$, homomesic with mean $|P|/h$ by [RW15] (the equality of the two means is also the 0-mesy of $T^+ - T^-$ of [DHPP23], valid on every finite poset); the third is a linear combination. The toggle size is $|\max I| + |\min(P \setminus I)| = \text{ac}(I) + \text{ac}(R(I))$, and a sum of homomesic statistics, one composed with the map, is homomesic with the sum of the means [DHPP23, Thm 3.20 and the closure properties stated there]. $\square$

Checked on all 42 boards (C); E_7 : $3/2, 3/2, 4$; E_6 : $4/3, 4/3, 10/3$; the D_5 spinor: $5/4, 5/4, 5/2$. **Remark 7.2.** On every non-chain board the number of distinct colours toggled, the per-weight kept count of Proposition 4.1, and the number of lattice elements below a weight are *not* homomesic; on the 56 the ternary counts are not homomesic under pr, and are homomesic under the sheet clock R_6 on each 27-sheet but not on the 56 (the poles are fixed points).

8. The computational model

The objects of §§3–7 were first met inside an explicit model, built in plain Python from generators in one coordinate system, of the following classical chain. We record what was constructed and how each identification was made; nothing in this section is claimed as new group theory.

8.1 The roof. $W(E_8)$ has order 696,729,600; its rotation subgroup $W^+(E_8)$ has index 2 and $W^+(E_8)/\{\pm 1\} \cong O_8^+(2)$, simple of order 174,182,400 [Con71, CS99, ATLAS]. $W(E_7) \cong 2 \times Sp_6(2)$, and $Sp_6(2)$ acts on the 28 bitangents of a plane quartic [Dol12]. The 56 weights of the minuscule representation of E_7 are the roots r of E_8 with $\langle r, \alpha \rangle = 1$ for a fixed root α ; the E_7 root system is $\alpha^\perp$ [Bou02].

Proposition 8.1 (C). *Two families of finite groups — the chain $PSL(2, 7) \supset C_6 \times C_2, A_5, S_4, A_4, C_2$ and the tower $SL(2, 7), 2 \cdot I = SL(2, 5), GL(2, 3), 2 \cdot O, 2 \cdot T = SL(2, 3), C_3 \times D_4, 2 \cdot W(E_6), 2 \times PGL(2, 7)$ together with the split covers $A_5 \times C_2, A_4 \times C_2$ — are exhibited by explicit witnesses as subgroups of $W(E_8)$ acting on the 240 roots, the first inside a vector-type subgroup $C \cong Sp_6(2)$, the second inside $H \cong 2 \cdot Sp_6(2)$. The group $K = \langle C, H \rangle$ has order 348,364,800 = $|W(E_8)|/2$ and every generator has determinant $+1$ (exact Bareiss elimination), so $K = W^+(E_8)$;*

K is perfect, and the 21 witness generators alone regenerate it. The centre of H is $\{\pm 1\}$: both families hang over a single central C_2 . (44 checks.)

8.2 Triality and the still point. $\text{Out}(O_8^+(2)) \cong S_3$; the fixed group of a triality automorphism of order 3 is $G_2(2) \cong U_3(3):2$ of order 12,096 [Con71, Kle87, Wil09, §4.7].

Proposition 8.2 (C). *Let $\bar{C} \cong Sp_6(2)$ be the image of C in $\Omega = O_8^+(2)$ (its vector shadow) and τ' the outer automorphism of Ω realised in the model. The intersection $\bar{C} \cap \tau'(\bar{C})$, computed by sifting all 1,451,520 elements of $\bar{C}$ through a base and strong generating set of its image, has order 12,096, derived subgroup of order 6,048 and element orders $\{1, 2, 3, 4, 6, 7, 8, 12\}$: it is $G_2(2)$. There is an intertwiner y , unique up to a 2-dimensional solution space, with $\tau'' = \text{inn}(y^{-1}) \circ \tau'$ of exact order 3 fixing this $G_2(2)$ pointwise. Inside it, $PSL(2, 7)$ occurs as a single class of 36 subgroups, each with normaliser $PGL(2, 7)$, all of the class transitive on the 28 bitangents; the still point lifts split to $2 \times G_2(2)$ in the spin cover. (19 checks; the split lift contradicted our expectation of a twisted lift.)*

Proposition 8.3 (C). *The odd half-spin module S^- carries a unique quadratic form Q^- of plus type; letting $\Omega = O_8^+(2)$ act on the $360 = 3 \times 120$ nonsingular points of $V \oplus S^+ \oplus S^-$ and defining Φ blockwise (T''^{-1} on V , where T'' is the matrix inducing τ'' on V ; the unique intertwiner $S^+ \rightarrow S^-$, the closing map on S^-), Φ has exact order 3, conjugates the generators as τ'' does, and $|\langle \Omega, \Phi \rangle| = 522,547,200 = 3 |O_8^+(2)|$: this is $O_8^+(2):3$ as an explicit permutation group on 360 points. The still point is the pointwise stabilizer of the Φ -orbit $\{v, \Phi v, \Phi^2 v\}$ — one vector, one spinor, one co-spinor. (16 checks.)*

The composition of Φ with a bridge element of order 7 has order 21, and with the image of a Coxeter element of $W(E_8)$ has order 18: these are the outer classes 21A and 18A/B of §8.4. The number 18 is the Coxeter number of E_7 ; its appearance here is a fact about $O_8^+(2):3$, distinct from Rush–Shi’s theorem on the board.

8.3 The board under the still point.

Proposition 8.4 (C). *The 56 weights Λ are the 56 nonsingular vectors u with $B(u, v) = 1$ for a fixed nonsingular v , equivalently the 56 roots of E_8 with $\langle r, \alpha \rangle = 1$, matched bijectively by Dynkin labels with all 84 covering relations of the weight lattice equal to subtraction of a simple root. Under this identification $\iota = -1$ is the transvection $t_v : u \mapsto u + v$ (Dickson invariant 1, outside Ω), while R and pr admit no linear extension. The still point $G_2(2)$ is transitive on the 56 weights, on the 28 bitangents, on the 36 even theta characteristics and on the 63 nonzero vectors of $\mathbb{F}_2^6$, with stabilizers of orders 216, 432, 336, 192; each even theta characteristic’s stabilizer is a $PGL(2, 7)$ whose derived group is one of the 36 subgroups of Proposition 8.2, bijectively. (14 checks.)*

The fraction of weight pairs whose inner-product type is preserved is 100% for ι and for every Weyl element, 93.0% for pr and 65.1% for R (Proposition 4.1) — the observation from which §§3–4 grew.

8.4 The fourteen outer classes. The conjugacy classes of $O_8^+(2):3$ in the outer coset $\Omega\Phi$ are classical data [ATLAS; Bre, CTblLib 1.3.11]. The model re-derives them by centralizer closures from the explicit permutation group of Proposition 8.3:

order	classes	centralizer orders in Ω	density
24	2	8, 8	$1/8 + 1/8$
21	1	7	$1/7$
18	1	6	$1/6$
12	4	32, 4, 48, 96	$5/16$
9	1	18	$1/18$
6	3	192, 24, 48	$13/192$
3	2	12,096 (the turn), 216	$1/12,096 + 1/216$

The densities sum to 1 exactly, so there is no fifteenth class **(C)**. The turn’s centralizer in Ω is exactly $G_2(2)$; the element of order 18 has centralizer of order 6, and its sixth power has centralizer of order 1,944 with centre of order 9. Against the GAP Character Table Library’s table of $O_8^+(2):3$ (headers only, no character values used) the outer class count 14 is forced by the published class numbers (53 for Ω , 55 for $\Omega.3$, a fusion fixing 14 classes and fusing 13 triples); the multiset of (order, centralizer order) of the published outer classes equals ours tripled;

all sixteen recorded power-map entries agree; the sixth power of the eighteen is class $3D$ of $O_8^+(2)$, the E_8 and D_4 Coxeter hearts are the fused triple $3A/3B/3C$, and the two classes of order 24 are told apart by their squares exactly as the table tells them apart **(C)**.

9. The bitangent bridge

9.1 $PSL(2, 7)$. $PSL(2, 7) \cong GL(3, \mathbb{F}_2)$, order 168, classes $1A, 2A, 3A, 4A, 7A, 7B$ of sizes 1, 21, 56, 42, 24, 24, character table

	1A	2A	3A	4A	7A	7B
χ_1	1	1	1	1	1	1
χ_3	3	-1	0	1	α	$\bar{\alpha}$
$\bar{\chi}_3$	3	-1	0	1	$\bar{\alpha}$	α
χ_6	6	2	0	0	-1	-1
χ_7	7	-1	1	-1	0	0
χ_8	8	0	-1	0	1	1

with $\alpha = (-1 + \sqrt{-7})/2$ (re-derived by the Burnside–Dixon class-algebra method and verified exactly over $\mathbb{Q}(\sqrt{-7})$ **(C)**). Since $7 \nmid |W(E_6)| = 51,840$ and $168/27 \notin \mathbb{Z}$, $PSL(2, 7)$ embeds in no $W(E_6)$ -geometry and has no transitive action on 27 points. At field level, $\text{Gal}(\mathbb{Q}(\zeta_{21})/\mathbb{Q}) \cong C_6 \times C_2$ has quadratic subfields $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{-7})$, $\mathbb{Q}(\sqrt{21})$; $\mathbb{Q}(\sqrt{-7})$ is the character field of $PSL(2, 7)$ and the CM field of the Klein quartic [Elk99], $\mathbb{Q}(\sqrt{-3})$ the Eisenstein field of E_6 ; they are linearly disjoint.

9.2 Theta characteristics. Realise $PSL(2, 7) = GL(3, \mathbb{F}_2)$ on $V \oplus V^* \cong \mathbb{F}_2^6$ with the symplectic form

$$\omega((v, f), (w, h)) = f(w) + h(v).$$

The 64 quadratic refinements of ω split by Arf invariant into 36 even and 28 odd; $Sp_6(2)$ acts on them with the stabilizer of an even form $O_6^+(2) \cong S_8$ and of an odd form $O_6^-(2) \cong W(E_6)$ [Dol12, ch. 6].

Proposition 9.1 (C). (i) $PSL(2, 7)$ fixes exactly one even refinement; its orbits on the 36 even refinements have sizes 1, 7, 7, 21, and it sits inside the stabilizer S_8 of the fixed form by its action on $\mathbb{P}^1(\mathbb{F}_7)$. (ii) $PSL(2, 7)$ is transitive on the 28 odd refinements with stabilizer $S_3 = N(\langle z_3 \rangle)$, the normaliser of a subgroup of order 3, and permutation character $\chi_1 + 2\chi_6 + \chi_7 + \chi_8$. (iii) Inside $Sp_6(2) \cong W(E_7)/\{\pm 1\}$: $W(E_6)$ is the stabilizer of one odd refinement (one bitangent), $PSL(2, 7)$ is transitive on all 28, and $PSL(2, 7) \cap \text{Stab} = S_3$. (iv) $C_{Sp_6(2)}(PSL(2, 7)) = 1$ and $N_{Sp_6(2)}(PSL(2, 7)) = PGL(2, 7)$, the outer automorphism realised as the symplectic duality $X \leftrightarrow X^*$ (by an exhaustive pass over all 1,451,520 elements, and independently by a structural argument). (v) The map $z \mapsto N(\langle z \rangle)$ is exactly 2-to-1 from the 56 elements of class $3A$ onto the 28 bitangent stabilizers, and the lift of $PSL(2, 7)$ to $W(E_7)$ acts on the 56 weights with two orbits of size 28, interchanged by ι , with weight stabilizer S_3 ; the two lifts to $W(E_7) = 2 \times Sp_6(2)$ coincide because $PSL(2, 7)$ is perfect.

The S_3 of (iii) is the stabilizer both of a bitangent in $W(E_6)$'s geometry and of a point in $PSL(2, 7)$'s action on 28 points — the two subgroups are the two points of view on one configuration. Two canonical restrictions of the 27-dimensional representation of E_6 follow from the two natural embeddings: through $E_6 \supset SU(3)^3$ with $PSL(2, 7)$ diagonal by χ_3 , $27|_{PSL(2,7)} = 3\chi_1 \oplus 3\chi_8$; through $PSL(2, 7) \subset G_2 = \text{Aut}(\mathbb{O})$ with $27 = J_3(\mathbb{O})$, $27| = 6\chi_1 \oplus 3\chi_7$ **(C)**; no restriction of the 27 contains a 16-dimensional constituent.

9.3 The Clifford reading. $Sp_6(2) = W(E_7)/\{\pm 1\}$ is the 3-qubit Clifford group modulo phases and Paulis; in that dictionary the 28 bitangents are the 28 odd Pauli sign-functions, $W(E_6)$ the Clifford stabilizer of one, and $PSL(2, 7)$ a Clifford subgroup transitive on all 28 (exhibited computationally in shared coordinates, 35 checks **(C)**). R has no image in this quotient: $\iota R \iota = R^{-1}$ while ι is central, and R does not act on the 28 antipodal pairs (Corollary 5.2(iii)) — the group-theoretic face of Theorem A.

9.4 The sextet and the Fano plane.

Proposition 9.2. χ_6 is the deleted permutation module on the 7 points of the Fano plane: realising $PSL(2, 7) = GL(3, \mathbb{F}_2)$ on the 7 nonzero vectors of $\mathbb{F}_2^3$, generators A (order 2), B (order 3) with AB of order 7 have traces 2, 0, -1 on the 6-dimensional sum-zero subspace, which is χ_6 's character at $2A, 3A, 7A/7B$ (cf. [LNR07, §7]). The two orbits of size 7 in Proposition 9.1(i) both carry permutation character $1 \oplus \chi_6$ (fixed-point counts 7, 3, 1, 1, 0, 0): they are the points and the lines of one Fano plane, dual under the outer automorphism, each realising χ_6 as its deleted permutation module. **(C)**

Consequently $\mathbb{C}\Lambda = 2(\chi_1 + 2\chi_6 + \chi_7 + \chi_8)$ contains the Fano plane's geometry four times over and the three-dimensional representations not at all; the octet χ_8 is the Steinberg representation, and the 2-transitive action of $PGL(2, 7)$ on $\mathbb{P}^1(\mathbb{F}_7)$ appears as the stabilizer S_8 of the fixed even theta characteristic.

10. Open problems

1. **Closed forms for κ** on the half-spins, on E_6 and E_7 , and on $A_n\omega_k$ for $k \geq 3$ (Remark 3.6 suggests degree $2k - 2$ in n).
2. **The centralizer law.** Proposition 5.4 is a computation on 23 boards. Is $I_k = C_W(R^k)$ on every simply-laced minuscule board on which no power of R is a Weyl element, apart from the D_5 half-spin? On the boards with a W -central reverser, Lemma 5.6 gives $I_k \subseteq C_W(R^{2k})$, with equality on the vector boards (Theorem 5.7); the inclusion is strict on $A_5\omega_3, A_7\omega_4$ and the D_6 half-spins at the half-turn. Is there a description of I_k inside $C_W(R^{2k})$ on those boards?
3. **The spinor boards.** Among $B_3, \dots, B_8$ the shared grammar exceeds the centralizer only on B_4 and, through $W(D_4)$, on B_3 (Proposition 6.7). Is B_4 the only B_n spinor board with $n \geq 4$ on which it does? The clock is free exactly when n is a power of 2, and B_{16} (65,536 weights) is the next free case; on B_{16} the correction's centralizer is again $\{\pm 1\}$ (Proposition 6.11), and Theorem 6.14 shows that $-(e_2e_3)(e_4e_5)\cdots$ commutes with $R^{h/2}$ only at $n = 4$. What remains is to show that this element is the Weyl element nearest to the half-turn for every $n > 8$, and that $C_W(R^{h/2}) = \{\pm 1\}$ for every $n \geq 5$ (computed for $n \leq 8$ and $n = 16$).
4. **Non-simply-laced and non-minuscule boards**, promotion, and birational rowmotion [Oka21]: none is treated here.
5. $\langle W, R \rangle$ on the boards where it could not be enumerated: it is A_{16} on the D_5 half-spin and A_{56} on E_7 ; "alternating" is an observation on five boards.
6. **The transports.** In which group below $\text{Sym}(2n)$, if any, are the outer transports of the odd D_n vector boards inner (Proposition 5.8)?

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Data and code availability

Every statement marked **(C)** is established by a named Python 3 script (numpy; sympy in one place) listed in Appendix B. The scripts, their hash-locked briefs, their logs including first runs, the witnesses they write, and the two editions of this paper are kept under version control at

https://github.com/SelinaAliens/Scar_merkabit

The repository is public under the MIT licence. Each script re-checks its brief's SHA-256 as its first test and runs from a clean checkout with no external data.

Declarations

Funding. This work received no external funding. *Competing interests.* The authors declare none. *Author contributions.* S.S. built the computational model and the verification framework, ran the computations, and drafted the paper. I.B. built the $PSL(2, 7)$ framework of §9, re-executed every computation independently, conducted the literature search, and reviewed the text. Both authors approved the final version.

Appendix A. The literature search

The claims of novelty in this paper are negative claims — “not found” — and are worth exactly the search behind them. The question: is it stated or proved anywhere that rowmotion on a minuscule poset, carried to the weights, is or is not an element of the Weyl group; for which posets it or its half-turn is; that a pair of weights keeps its inner product under rowmotion according to a toggle parity; or what fraction of pairs it keeps?

Keyword and citation-trail searches. Toggle groups, periodicity, homomesy, cyclic sieving, rowmotion with quadratic or bilinear forms mod 2, the E_7 case and the Gosset graph; and the thirty documents citing [RS13] on zbMATH Open, all read. Found: the Coxeter-motion and birational literature [Oka21], the toggle literature [SW12, Str18], the homomesy literature [PR15, RW15, DHPP23]; no match.

Corpus. The arXiv API query `all:rowmotion` returns the complete corpus of 57 documents (2014–2026). Every title and abstract was read. Titles containing *minuscule*: Hopkins 2016 (CDE property for minuscule lattices), Hopkins 2019 (minuscule doppelgängers), Okada [Oka21], Pechenik [Pec22]. Titles containing *Coxeter*: Okada [Oka21], Marczinzik–Thomas–Yıldırım [MTY24]. Titles containing *Weyl, isometry, orthogonal, inner product* or E_7 : none. The remaining titles concern Tamari and Cambrian lattices, fences, root posets and rowvacuation, interval-closed sets, products of chains, birational and piecewise-linear lifts, Markov chains, rooted trees, trapezoids, plane partitions, semidistributive and trim lattices, alternating sign matrices, and echelonmotion. The zbMATH Open API returned nine documents for the same term, none with these terms in title or keywords.

Primary texts, read for the question. [RS13]: Theorem 1.3, rowmotion and the toggle product $t_{(i_1, \dots, i_n)}$ are conjugate in the toggle group; Theorem 1.4, under Stembridge’s bijection $J(P) \rightarrow W^J$ that product corresponds to a Coxeter element; no statement that rowmotion itself is or is not a Coxeter element’s action; the chain case attributed to Stanley [Sta09]; inner products, isometries and pairs of weights not mentioned. [Oka21]: Coxeter-motion defined as a product of file toggles and shown conjugate to rowmotion in the birational toggle group (Thm 4.3) — two maps, conjugate, not equal; the double-tailed diamond $D_n \omega_1$ treated birationally with separate formulas for the two tails, without any statement about Weyl group membership. [Hop24]: records the Rush–Shi conjugacy and the open conjectures on cyclic sieving and doppelgängers; nothing on Weyl membership or pairs. [TW19]: trim lattices; nothing on the weights.

Adjacent results, not the same statement. [Oka21] and [RS13] both relate rowmotion to a Coxeter-type action by conjugacy and neither asks when the conjugating element can be taken trivial (Theorems 3.2–3.3). [MTY24] study a homological “Coxeter matrix” built from the incidence algebra’s Cartan matrix alongside rowmotion and prove a different commutation identity. [Pan16] indexes lower ideals of weight posets by Weyl group elements, not rowmotion by one.

What the search does not reach. MathSciNet and the full text of zbMATH’s reviews; theses and work in progress. A reader who knows Theorems A–G as corollaries of the Rush–Shi conjugacy or of the toggle description of rowmotion is asked for the reference.

Appendix B. Verification

Every statement marked (C) is established by a Python 3 script (numpy; sympy in one place) whose brief was fixed and hashed before the script was written and which re-checks that hash as its first test. Each script was executed by both authors from clean checkouts on independent machines with identical tallies. The scripts, with their check counts and external inputs:

All scripts are at the root of the repository except `verify_tbr_bridge.py` and `verify_tsc_scarcat.py` (under package/) and `fano_sextet_six_invariants.py` (in the received-packages folder). “Inputs” are files written by the scripts of earlier rows; all are regenerable.

statement	script	checks	inputs
Prop. 8.1	<code>verify_stone_v_we8_roof44.py</code>	44	cache of
Prop. 8.2	<code>verify_stone_z_stillpoint19.py</code>	19	<code>verify_stone_u_2cover.py</code>
Prop. 8.3	<code>verify_stone_aa_roofclo16.py</code>	16	caches of the u, v, w, x scripts
Prop. 8.4	<code>verify_stone_ab_merkabi14.py</code>	14	the same, plus the cache of <code>verify_stone_z_stillpoint.py</code>
Prop. 4.1 (E ₇), §4.1	<code>verify_stone_ac_parity_9u12.py</code>	9, 12	<code>scar56_data.json</code> , <code>G_rows.npy</code>
§8.4 (14 classes)	<code>verify_stone_ai..am*.py</code>	15, 17, 14, 15+2, 15+1	<code>scar56_data.json</code> caches of the aa and z scripts
§8.4 (ATLAS names)	<code>verify_stone_an_atlas_n17es.py</code>	17	+ two CTblLib header files
Prop. 4.4	<code>verify_stone_ao_mirror_11d1.py</code>	11, 1	<code>scar56_data.json</code>
Prop. 4.2	<code>verify_stone_ap_clock_c22tr2lizer.py</code>	22, 2	<code>scar56_data.json</code>
Table 1	<code>verify_stone_aq_rush_sh8+d2.py</code>	8, 2	none (Cartan matrices)
Thms 3.2–3.5 checks	<code>verify_stone_ar_nonlinear10ity_theorems.py</code>	10	the board machine of the aq script
Thm 4.3	<code>verify_stone_as_shared_g14ammar.py</code>	14	witnesses of the ap script, <code>scar56_data.json</code>
Prop. 5.4	<code>verify_stone_at_grammar7family.py</code>	7	the board machine of the aq script
Thm 5.1, Prop. 4.5	<code>verify_stone_au_two_rev10sals.py</code>	10	witnesses of the aq, ap scripts
§6.1	<code>verify_stone_av_d5_anom9ly4.py</code>	9, 4	none
Prop. 7.1	<code>verify_stone_aw_homomes6.py</code>	6	the board machine of the aq script
Props. 4.6, 5.3	<code>verify_stone_ax_c_equal21tr1.py</code>	21, 1	witnesses of the aq, at, ap, as, au scripts
Thm 6.2, Lemma 6.1	<code>verify_stone_ay_d5_by_h20d15.py</code>	20, 15	witnesses of the av, aq, at scripts
Thm 5.7, Prop. 5.8	<code>verify_stone_az_doubled11b2k.py</code>	11, 2	witnesses of the av, aq, at scripts
Prop. 6.6	<code>verify_stone_ba_clifford7_hierarchy.py</code>	7	witnesses of the aq, at, av scripts
Thm 6.2 (rank-shift)	<code>verify_stone_bb_halfturn6_fullheight.py</code>	6	none
Prop. 6.7	<code>verify_stone_bc_b8_half14m4.py</code>	14, 4	none
Lemma 6.9, Prop. 6.10	<code>verify_stone_bd_halfturn7+c3.py</code>	7, 3	none
Prop. 6.11	<code>verify_stone_be_b16_orb9us2.py</code>	9, 2	none
Lemma 6.12, Prop. 6.13, Thm 6.14	<code>verify_stone_bf_why_fou8.py</code>	8	none
Prop. 9.1	<code>verify_tbr_bridge.py</code> , 20, 11 <code>verify_stonef_b_dq1.py</code>	20, 11	none
§9.1 table	<code>verify_tsc_scarcat.py</code>	48	none
§9.3	<code>verify_stoneq_clifford.35</code>	35	none
Prop. 9.2	<code>fano_sextet_six_invariants.py</code> <code>verify_fano_bitangent_identity.py</code>	no tally	none

“Inverted” counts record expectations registered before the computation that the computation refuted; each is described where it is informative (Remarks 3.7, 5.5, 5.9; Propositions 8.2, 8.4). The scripts, the hash-locked briefs and the logs are in the repository named under *Data and code availability*.

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